SOMAIYA

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K J Somaiya Institute of Technology

An Autonomous Institute Permanently Affiliated to the University of Mumbai

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Class : Second Year (SY) 2026-27 **Department : Information Technology**

Course code: ITXS37 **Subject: DSA**

Experiment No:4

Aim: Experiment No. 04 (Implementation double ended queue using Array)

Program:

```
#include <stdio.h>



#define MAX 5



int deque[MAX];

int front = -1;

int rear = -1;



int isFull() {

    return ((front == 0 && rear == MAX - 1) || (front == rear + 1));
```

```
}



int isEmpty() {

    return (front == -1);

}




void insertFront(int value) {

    if (isFull()) {

        printf("Deque Overflow! Cannot insert %d at front.\n", value);

        return;

    }
```

```
    if (front == -1) { /* Empty deque */

        front = rear = 0;

    } else if (front == 0) {

        front = MAX - 1; /* Wrap around */

    } else {

        front = front - 1;

    }

    deque[front] = value;

    printf("Inserted %d at FRONT.\n", value);

}
```

```
void insertRear(int value) {

    if (isFull()) {

        printf("Deque Overflow! Cannot insert %d at rear.\n", value);

        return;

    }

    if (front == -1) { /* Empty deque */

        front = rear = 0;

    } else if (rear == MAX - 1) {

        rear = 0; /* Wrap around */

    } else {

        rear = rear + 1;

    }
```

```
    deque[rear] = value;

    printf("Inserted %d at REAR.\n", value);

}

void deleteFront() {

    if (isEmpty()) {

        printf("Deque Underflow! Cannot delete from front.\n");

        return;

    }

    printf("Deleted %d from FRONT.\n", deque[front]);

    if (front == rear) { /* Only one element */
```

```
        front = rear = -1;

    } else if (front == MAX - 1) {

        front = 0; /* Wrap around */

    } else {

        front = front + 1;

    }

}




void deleteRear() {

    if (isEmpty()) {

        printf("Deque Underflow! Cannot delete from rear.\n");
```

```
        return;

    }

    printf("Deleted %d from REAR.\n", deque[rear]);

    if (front == rear) { /* Only one element */

        front = rear = -1;

    } else if (rear == 0) {

        rear = MAX - 1; /* Wrap around */

    } else {

        rear = rear - 1;

    }

}
```

```
void display() {
    if (isEmpty()) {
        printf("Deque is empty.\n");
        return;
    }
    printf("Deque elements: ");
    int i = front;
    while (1) {
        printf("%d ", deque[i]);
        if (i == rear)
            break;
```

```
        i = (i + 1) % MAX;

    }

    printf("\n");

}




int main() {

    int choice, value;



    while (1) {

        printf("\n----- Deque Menu -----\n");

        printf("1. Insert at Front\n");
```

```
printf("2. Insert at Rear\n");

printf("3. Delete from Front\n");

printf("4. Delete from Rear\n");

printf("5. Display\n");

printf("6. Exit\n");

printf("Enter your choice: ");

scanf("%d", &choice);



switch (choice) {

    case 1:

        printf("Enter value: ");

        scanf("%d", &value);
```

```
            insertFront(value);
            break;
        case 2:
            printf("Enter value: ");
            scanf("%d", &value);
            insertRear(value);
            break;
        case 3:
            deleteFront();
            break;
        case 4:
            deleteRear();
```

```
            break;

        case 5:

            display();

            break;

        case 6:

            printf("Exiting...\n");

            return 0;

        default:

            printf("Invalid choice! Try again.\n");

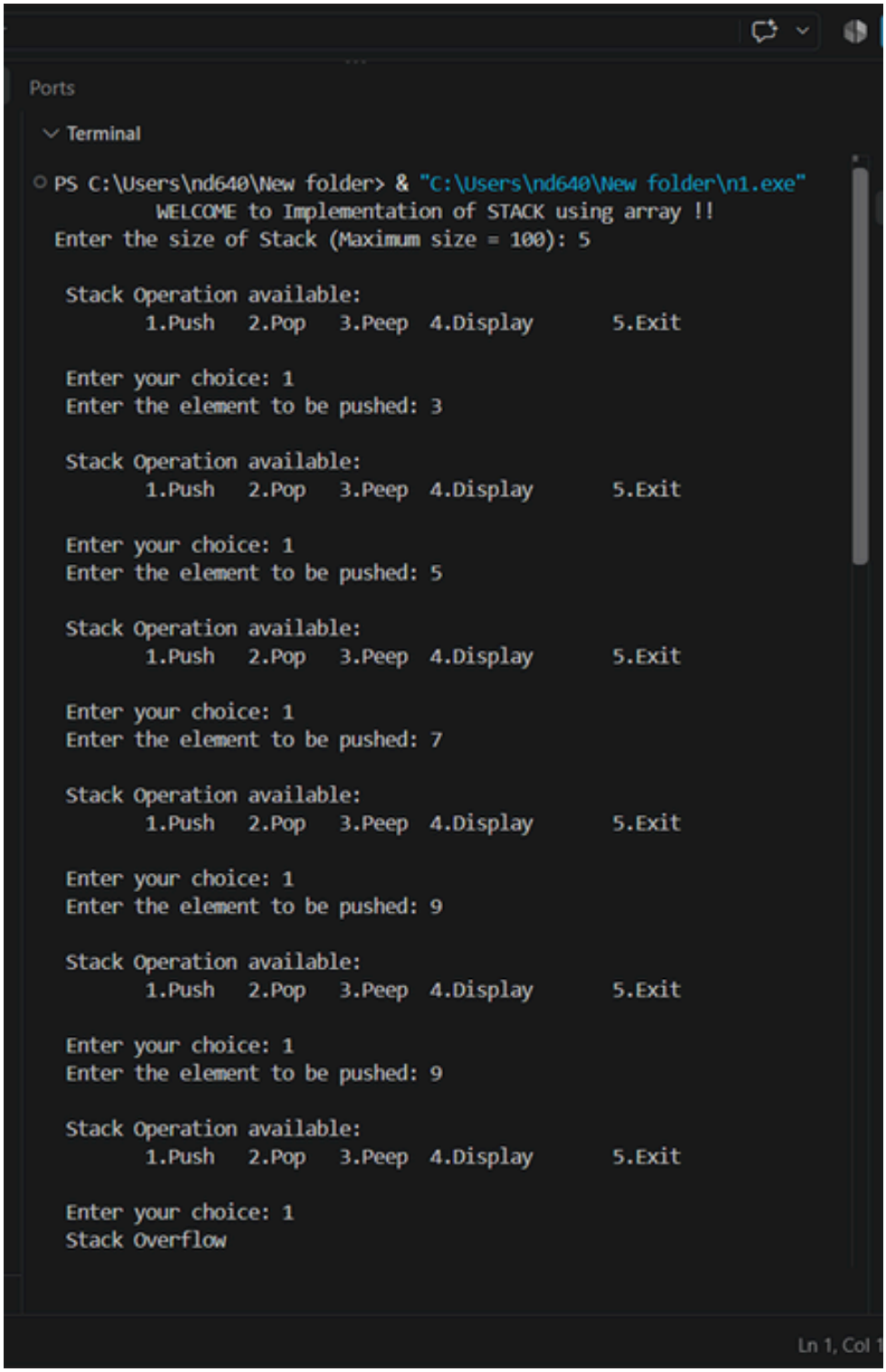
    }

}

return 0;
```

```
}
```

Outcome:


Ports
Terminal
PS C:\Users\nd640\New folder> & "C:\Users\nd640\New folder\n1.exe"
WELCOME to Implementation of STACK using array !!
Enter the size of Stack (Maximum size = 100): 5
Stack Operation available:
1.Push 2.Pop 3.Peep 4.Display 5.Exit
Enter your choice: 1
Enter the element to be pushed: 3
Stack Operation available:
1.Push 2.Pop 3.Peep 4.Display 5.Exit
Enter your choice: 1
Enter the element to be pushed: 5
Stack Operation available:
1.Push 2.Pop 3.Peep 4.Display 5.Exit
Enter your choice: 1
Enter the element to be pushed: 7
Stack Operation available:
1.Push 2.Pop 3.Peep 4.Display 5.Exit
Enter your choice: 1
Enter the element to be pushed: 9
Stack Operation available:
1.Push 2.Pop 3.Peep 4.Display 5.Exit
Enter your choice: 1
Enter the element to be pushed: 9
Stack Operation available:
1.Push 2.Pop 3.Peep 4.Display 5.Exit
Enter your choice: 1
Stack Overflow
Ln 1, Col 1

Conclusion: SUCESSFULLY EXCUTED THE DEQUEUE USING STRING